

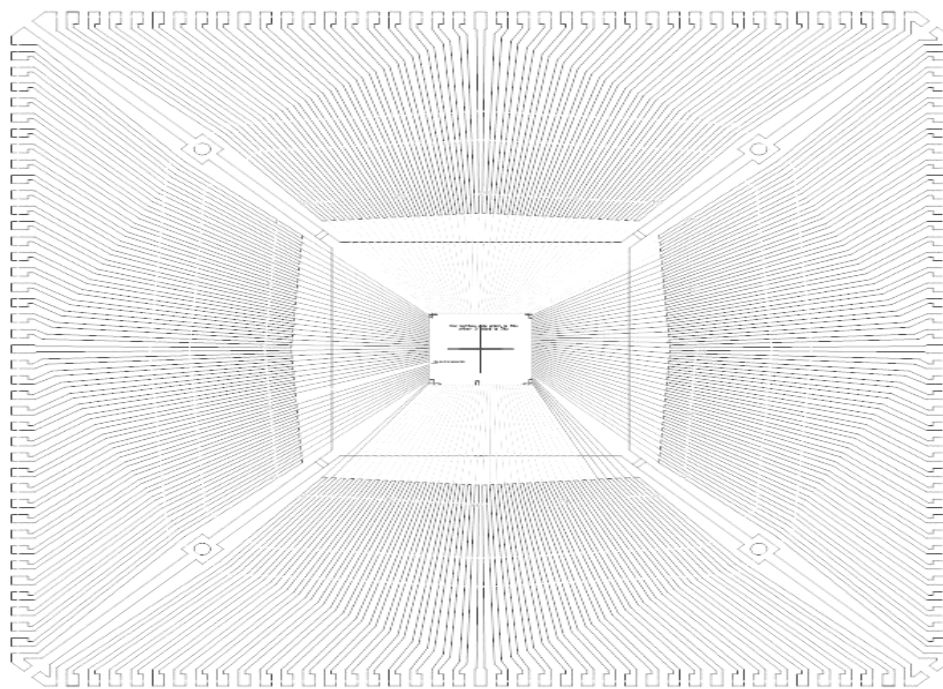
# LONTIUM SEMICONDUCTOR CORPORATION

ClearedEdge™ Technology

## LT6711GXE Board\_Design\_Guide

### For HD\_DVI Interface

龙迅半导体



**We produce mixed-signal products for a better digital world!**

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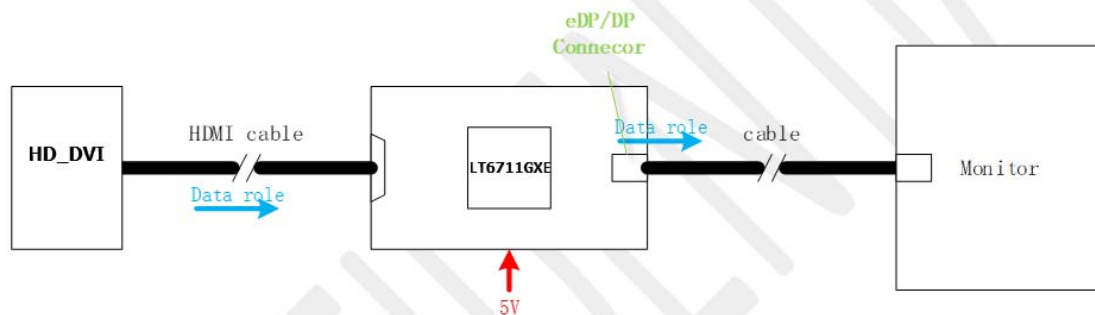
## 2 前言 (PREFACE)

LT6711GXE 是一颗 HD\_DVI2.1 to DP1.4 的转换 IC，支持内部 Type-C Alternate Mode switch 和 PD controller。

本文档详细描述在用 LT6711GXE 来设计 HD\_DVI 输入 DP 输出的应用下，关键部位的设计要求及参考，为了减小设计风险，请务必遵守。

(The LT6711GXE is HD\_DVI2.1 to DP1.4 converter with internal Type-C Alternate Mode switch and PD controller..

This document describes in detail the design requirements and key parts under the application of LT6711GXE to design HD\_DVI input and DP output. In order to reduce design risk, it is important to observe.)



### 3 原理图设计(SCHEMATIC DESIGN)

#### 3.1 HD\_DVI Input

LT6711GXE 支持 HD\_DVI 2.1 输入转 DP/eDP 输出。HD\_DVI DDC 信号 DDC\_SDA、DDC\_SCL 需要通过 47K 电阻上拉到 HD\_DVI\_5V。**RX\_HPD 请接 1K 电阻上拉到系统 5V。**HD\_DVI\_5V 请连接到 RX\_5V\_DET 做插入检测，若需要连接 SOC 做插入检测，请注意 SOC 电平是否匹配，若不匹配需要加电平转换；当检测到 HD\_DVI 插入，复位一下芯片再工作。**注意区分 HD\_DVI\_5V 和系统 5V。**

若 HD\_DVI 接口是标准的 HD\_DVI 连接器，请将 DDC\_SCL, DDC\_SDA, RX\_HPD 上拉到 5V。若 HD\_DVI 输入端信号来自上一级芯片，且在同一块板子上，DDC\_SCL、DDC\_SDA、RX\_HPD 请上拉到 3.3V 或者 5V，如果上一级芯片输出端口电平不兼容，请加电平转换电路。

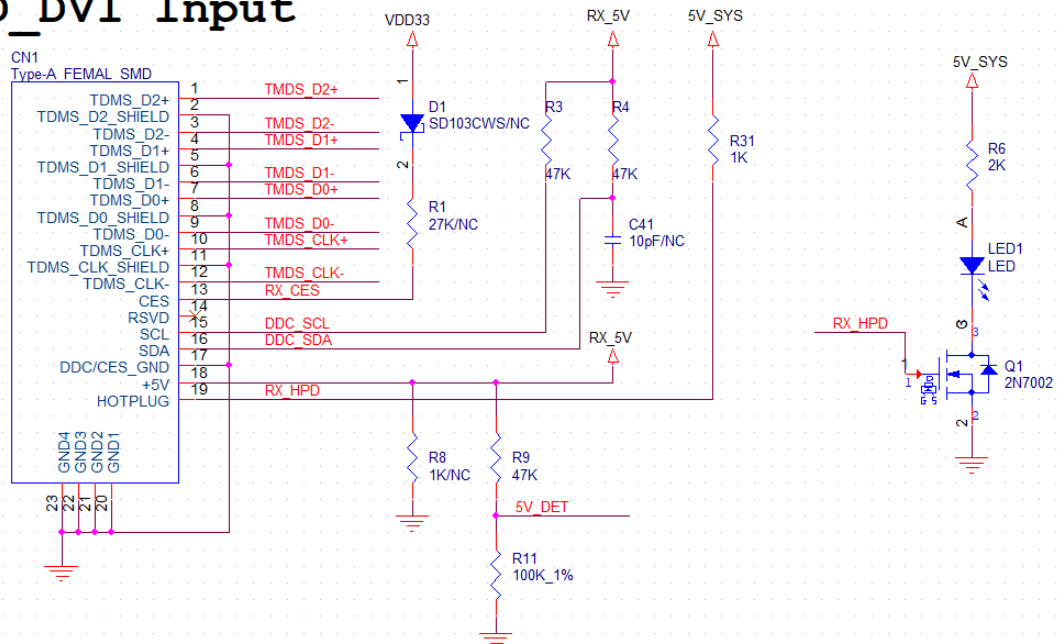
TMDS 信号建议加静电防护电路，DDC\_SCL、DDC\_SDA、RX\_HPD、HD\_DVI\_5V 同样做静电防护。

(The LT6711GXE supports HD\_DVI 2.1 input to DP/eDP output. The HD\_DVI DDC signals DDC\_SDA and DDC\_SCL need to be pulled up to HD\_DVI\_5V through a 47K resistor. **RX\_HPD Please connect a 1K resistor and pull it up to the system 5V.** Please connect HD\_DVI\_5V to RX\_5V\_DET for insertion detection. If you need to connect SOC for insertion detection, please pay attention to whether the SOC level matches. If it does not match, you need to add level conversion; when HD\_DVI insertion is detected, reset the chip and then work. **Note the distinction between HD\_DVI\_5V and system 5V.**

If the HD\_DVI interface is a standard HD\_DVI connector, please pull up DDC\_SCL, DDC\_SDA, RX\_HPD to 5V. If the HD\_DVI input signal comes from the upper-level chip and is on the same board, please pull up DDC\_SCL, DDC\_SDA, RX\_HPD to 3.3V or 5V. If the output port level of the upper-level chip is not compatible, please add a level conversion circuit .

It is recommended to add ESD protection circuit to TMDS signal, DDC\_SCL, DDC\_SDA, RX\_HPD , HD\_DVI\_5V also do ESD protection.)

## HD\_DVI Input



TMDS 信号默认只能做 DC Couple 输出;  
(TMDS signal can only be output by DC Couple by default)

TMDS 信号不支持 Lane Swap 和 Polarity Swap;  
(TMDS signals not support Lane Swap and Polarity Swap)

## 3.2 DP/eDP Output

DP 信号主要包括: DP\_PWR、AUX+/-、HPD 和 Main Link 信号。  
(The DP signals include DP\_PWR, AUX+/-, HPD and Main Link signals.)

### 3.2.1 Main Link 信号 (Main Link signal)

Main Link 信号一共有 4 对差分信号:

D0+/--(Lane0)、D1+/--(Lane1)、D2+/--(Lane2)、D3+/--(Lane3);

LT6711GXE 输出 DP 有三种方式: 1Lane 输出(Lane0), 2lane 输出(Lane0-1)和 4Lane(Lane0-3) 输出;

对应的速率有四种: 1.62Gbps、2.7Gbps、5.4Gbps、8.1Gbps;

不同的 Lane 和不同的速率, 对应所支持的最高分辨率也不同。

**Main Link 信号线序需要和参考设计保持一致, 不能更改**

(There are four pairs of differential signals in the Main Link :

D0+/- (Lane0), D1+/- (Lane1), D2+/- (Lane2), D3+/- (Lane3),

Main Link have three ways to output from LT6711GXE:

1Lane output(Lane0), 2Lane output(Lane0-1) and 4Lane output(Lane0-3).

Main Link have four rates: 1.62Gbps, 2.7Gbps, 5.4Gbps, 8.1Gbps.

The Lane count and rates is different, the maximum resolution of LT6711GXE output is different.

The Main Link signal line sequence needs to be consistent with the reference design and cannot be changed.)

Main Link 为 AC couple。按照 DP spec 的描述，Main Link 通路中的 AC 电容是加在 source 端，具体如下图所示：

(Main Link is AC couple. According to the description of DP spec, the AC capacitance in the Main Link path is added to the source, as shown in the figure below):

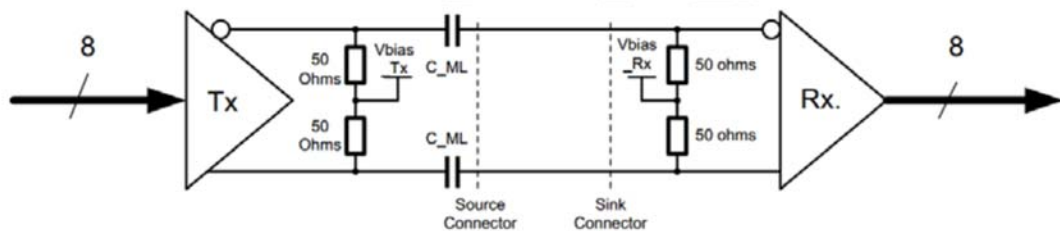


Figure 3-34: Main Link Differential Pair

此方案中我们 LT6711GXE 作为 Source，需要串 100nF 电容。

(We use LT6711GXE as a DP source and need a series of 100nF capacitors.)

### 3.2.2 AUX 信号 (AUX Signal)

AUX 信号是 AC couple。按照 Spec 规定，AUX 信号在 Source 端和 Sink 端都需要串联 0.1uF 电容，并且在两组电容之间需要有上下拉电阻：

(The AUX signal is AC couple. According to Spec regulations, the AUX signal needs to series with 0.1uF capacitors at both the Source and Sink terminals, and there needs to be a pull-up and pull-down resistor between the two sets of capacitors: )

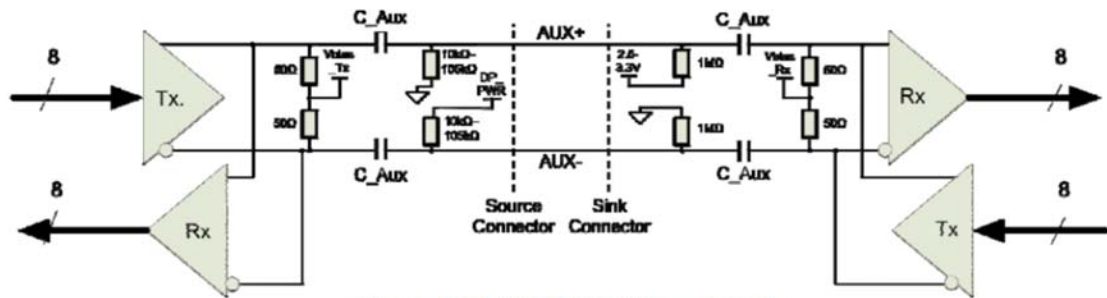


Figure 3-22: AUX CH Differential Pair

AUX 有检测 DP Source 插入的功能，如上图所示，当 DP Source 没有插入的时候，AUX+被 1M 上拉到 3.3V，电压为 3.3V，同时 AUX-被 1M 下拉到地，电压为 0V；当 DP source 插入后，AUX+被 1M 电阻和 100K 电阻分压，电压由 3.3V 变为 0.3V，同理 AUX-电压由 0V 变为 3V，这些变化可以作为 DP Source 插入检测。

(AUX has the function of detecting the insertion of DP Source. As shown in the figure above, when DP Source is not inserted, AUX + is pulled up to 3.3V by 1M and the voltage is 3.3V, while AUX - is pulled down to ground by 1M, the voltage is 0V. When DP source is inserted, AUX + is divided by 1M resistance and 100K resistance, the voltage is changed from 3.3V to 0.3V, and the AUX-voltage is also changed from 0V to 3V. These changes can be used as DP Source insertion detection.)

注：EDP 应用时，一般没有插拔动作，此时 AUX 信号上面可以不加上下拉电阻。Spec 规定如下：

(Notes: When you application is EDP output, it no plug action. At this time, the pull up/down resistor can be delete of the Aux signal. Spec as below: )

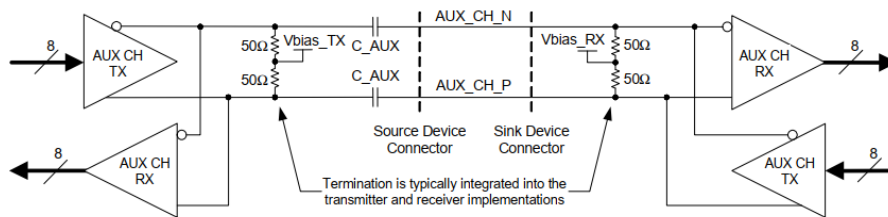


Figure 3-1: Recommended eDP AUX CH Topology

因此 EDP 应用时，100K 上下拉电阻可以不上件。

(When you application is EDP output, 100K pull-down resistor can be left unused)

### 3.2.3 TX\_HPD 信号(TX\_HPD Signal)

TX\_HPD: 热插拔信号。在下游有显示器接入时, 会给 LT6711GXE 高电平。  
LT6711GXE 的 TX\_HPD 信号不能有外部上拉, 需要在外部加 100K 下拉电阻。

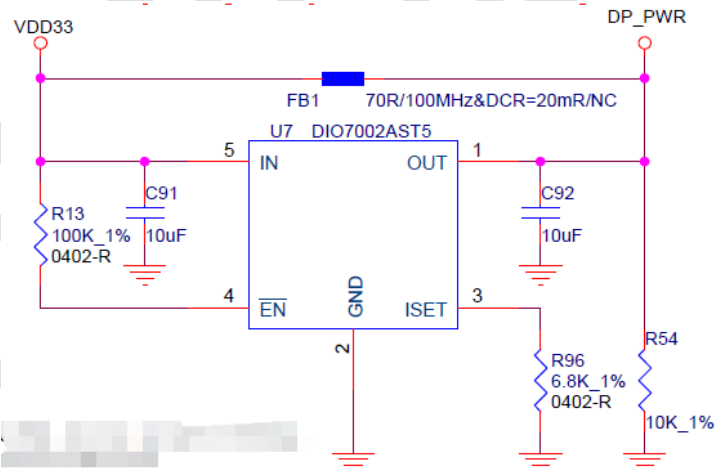
(TX\_HPD: When the downstream display is inserted , the display will give LT6711GXE a high level through HPD signal.

The TX\_HPD signal of LT6711GXE cannot have an external pull-up, and needs to add a 100K pull-down resistor externally.)

### 3.2.4 DP\_PWR 信号 (DP\_PWR Signal)

目前参考设计上面的是使用限流开关的方式去输出 DP\_PWR, 用限流芯片目的是为了过 CTS 认证。如果没有 CTS 认证需求的话, 可以将此部分电路更换为电源磁珠的方式处理。DP\_PWR 电路, 如下图所示:

(The current reference design uses a current-limiting switch to output DP\_PWR. The purpose of using a current-limiting chip is to pass CTS certification. If there is no CTS certification requirement, this part of the circuit can be replaced by a magnetic bead. DP\_PWR circuit, as shown in the figure below):



### 3.3 Crystal

LT6711GXE 晶振使用 25MHZ 无源晶体，建议选择负载电容小的晶体，频差在  $\pm 50\text{ppm}$  以内，要求晶体的  $G_{\text{mcrit}} \leq 2 \text{ mA/V}$ ，ESR:Equivalent Resistance,  $C_0$ :Shunt Capacitance,  $C_L$ :Load Capacitance，可在晶振数据手册中查找。

晶振 1M 负反馈电阻不能删除。

$$G_{\text{mcrit}} = 4 * \text{ESR} * (2\pi F)^2 * (C_0 + C_L)^2$$

$C_L$  并非 PCB 上焊接的真正负载电容，PCB 实际焊接负载电容计算公式如下：

$$C_{\text{Load}} = \left( \frac{[C_{\text{in}} + C_{L2}] \times [C_{L1} + C_{\text{out}}]}{[C_{\text{in}} + C_{L2} + C_{L1} + C_{\text{out}}]} \right) + C_{\text{PCB\_Stray}}$$

$C_{L1}$  和  $C_{L2}$  是 PCB 上要安装的负载电容， $C_{\text{PCB\_Stray}}$ :PCB 走线寄生电容约 3pF， $C_{\text{in}}$  和  $C_{\text{out}}$ :封装寄生电容约为 1pF。最终需要  $C_{\text{Load}}$  符合晶振数据手册要求。

(The LT6711GXE crystal oscillator uses a 25MHZ passive crystal. It is recommended to choose a crystal with a small load capacitance, the frequency difference is within  $\pm 50\text{ppm}$ , and the  $G_{\text{mcrit}}$  of the crystal is required to be less than or equal to 2 mA/V. Look in the crystal oscillator data sheet.

Crystal 1M negative feedback resistor can not be deleted.

$$g_{\text{mcrit}} = 4 \times \text{ESR} \times (2\pi F)^2 \times (C_0 + C_L)^2$$

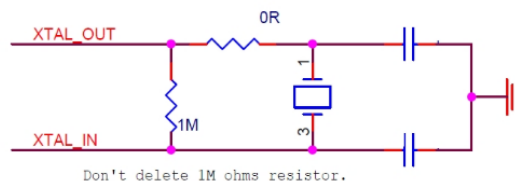
$$\text{ESR} = R_s \left( 1 + \frac{C_0^2}{C_L} \right)$$

ESR= Equivalent resistance of the crystal at load resonance  
 $C_0$  = Shunt capacitance of the crystal  
 $C_L$  = Equivalent load capacitance of the oscillation circuit

计算晶振负载电容：

Calculate the crystal load capacitance:

$$C_{\text{Load}} = \left( \frac{[C_{\text{in}} + C_{L2}] \times [C_{L1} + C_{\text{out}}]}{[C_{\text{in}} + C_{L2} + C_{L1} + C_{\text{out}}]} \right) + C_{\text{PCB\_Stray}}$$



CL1 和 CL2 是 PCB 上要安装的负载电容，公式计算后，让 Cload 符合晶振数据手册要求。

CPCB\_Stray: PCB 走线寄生电容（3pF），Cin 和 Cout: LT6711GXE 封装寄生电容（1pF）。

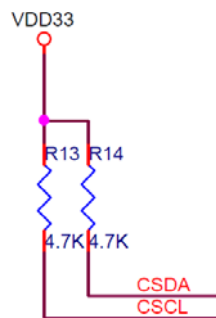
（CL1 and CL2 are the load capacitors to be installed on the PCB. The formula is calculated so that Cload meets the crystal data sheet requirements.

CPCB\_Stray: PCB alignment parasitic capacitance (3pF), Cin and Cout: LT6711GXE package parasitic capacitance (1pF). )

### 3.4 I2C(CSCL/CSDA)

CSCL/CSDA 为 I2C 接口，用于芯片固件升级和调试使用。推荐使用 100KHz 速率。LT6711GXE 内置 MCU，可预留 I2C 烧录口通过 I2C 烧录工具进行固件烧录和升级；对于系统应用，建议将 I2C 连接到 SOC 的 I2C 总线，通过 SOC 的 I2C 总线进行固件升级。**I2C 兼容 3.3V/5V 电平，请将 I2C 加 4.7K 电阻上拉（OD）。**建议预留 I2C 测试点，便于后期调试使用。

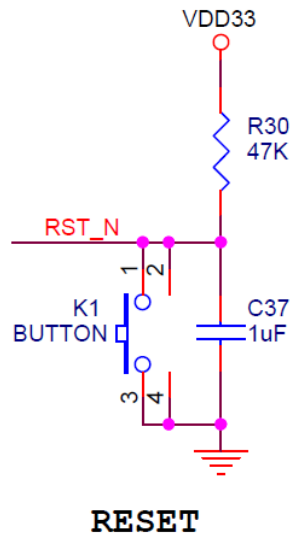
（CSCL/CSDA is an I2C interface, which is used for chip firmware upgrade and debugging. A rate of 100KHz is recommended. The LT6711GXE has a built-in MCU, which can reserve the I2C burning port for firmware burning and upgrading through the I2C burning tool; for system applications, it is recommended to connect the I2C to the I2C bus of the SOC, and upgrade the firmware through the I2C bus of the SOC. **I2C is compatible with 3.3V/5V level, please add 4.7K resistor to I2C to pull up (OD).** It is recommended to reserve I2C test points for later debugging.



### 3.5 RST\_N

芯片 RST\_N 引脚请加 47K 电阻上拉至 3.3V，并加 1uF 电容连接到 GND。强烈建议将 RST\_N 连接到 SOC 的 IO 进行控制，在 1.1V 和 3.3V 上电完成之后复位一下芯片再工作。

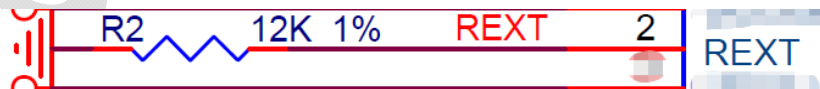
(Please add a 47K resistor to pull up the RST\_N pin to 3.3V, and add a 1uF capacitor to connect to GND. It is strongly recommended to connect RST\_N to the IO of the SOC for control, and reset the chip to work after 1.1V and 3.3V are powered on.)



### 3.6 REXT

REXT 务必连接一颗 12K\_1%精度电阻到 GND，并在 layout 时靠近芯片引脚放置，走线适当加粗。该引脚对应芯片内部的 BandGap 电路，需提供精确的参考电流。

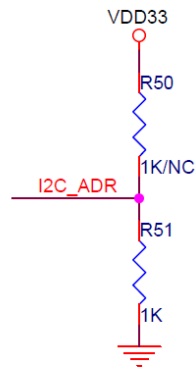
(REXT must be connected to a 12K\_1% precision resistor to GND, and placed close to the chip pins during layout, and the traces should be appropriately thickened. This pin corresponds to the BandGap circuit inside the chip and needs to provide an accurate reference current.)



### 3.7 I2C\_ADR

I2C\_ADR 为芯片地址选择引脚。可通过选择上拉或下拉电阻及浮空来选择合适的 I2C 地址。当使用 1K 电阻上拉时，地址为 0x9a；1K 电阻下拉时，地址为 0x86；20K 电阻下拉时，地址为 0x82；浮空时，地址为 0x9e。默认选择 1K 电阻下拉地址 0x86。

(I2C\_ADR is the chip address selection pin. Different I2C addresses can be selected by selecting pull-up or pull-down resistors and floating. When the 1K resistor is used to pull up, the address is 0x9a; when the 1K resistor is pulled down, the address is 0x86; when the 20K resistor is pulled down, the address is 0x82; when it is floating, the address is 0x9e. The 1K resistor pull-down address 0x86 is selected by default.)



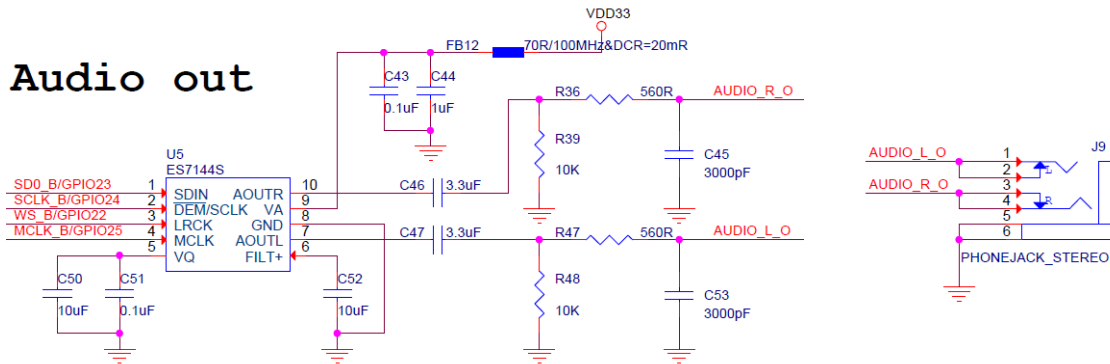
| I2C Address    |      |
|----------------|------|
| 1K pull-down   | 0x86 |
| 20K pull-down  | 0x82 |
| Floating       | 0x9e |
| 1K pull-up     | 0x9a |
| Default : 0x86 |      |

### 3.8 I2S/SPDIF

LT6711GXE 提供 I2S 和 SPDIF 两个数字音频输出接口。I2S 接口支持 8 通道 LPCM，采样率为 32~192 kHz。SPDIF 接口支持 LPCM，Dolby Digital，DTS 数字音频，最大采样率为 192kHz。(若不需要 IIS 功能，请将该 PIN 直接悬空。)

(The LT6711GXE provides two digital audio output interfaces, I2S and SPDIF. The I2S interface supports 8-channel LPCM with a sampling rate of 32~192 kHz. The SPDIF interface supports LPCM, Dolby Digital, DTS digital audio, and the maximum sampling rate is 192kHz. (If you don't need IIS functionality, please just leave this PIN hanging.))

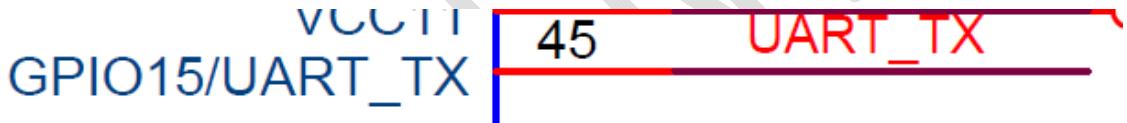
## Audio out



### 3.9 GPIO15/UART\_TX

GPIO15/UART\_TX 作为 UART\_TX 使用，请预留测试点，用于测试打印 LOG 使用，**波特率为 62500**。

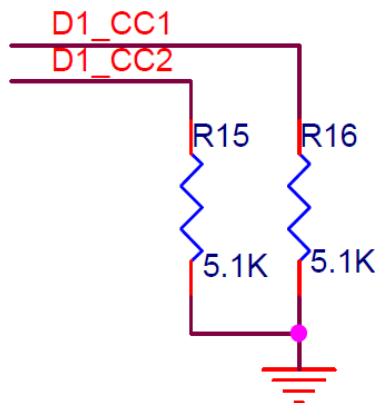
(GPIO15/UART\_TX is used as UART\_TX, please reserve a test point for test print LOG use, **baud rate is 62500**.)



### 3.10 D1\_CC1/D1\_CC2

当 DP 输出时，CC 务必要加 5.1K 下拉。

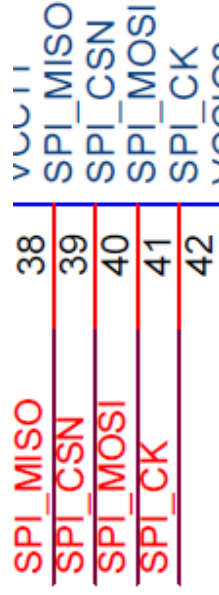
(For DP output, D1\_CC1/D1\_CC2 must be pull-down by 5.1K)



### 3.11 SPI

SPI 接口暂时未使用，**可以将该 PIN 直接浮空**。

(The SPI interface is not in use for the time being, so you can just float that PIN.)



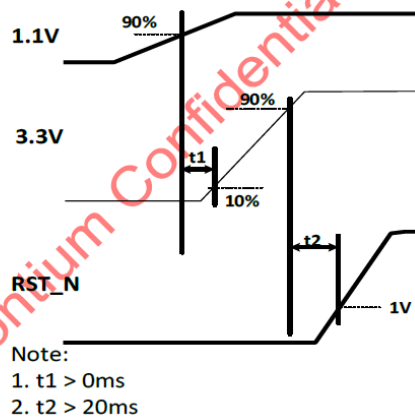
### 3.12 Power On and Power off Sequence:

如果你有外部 MCU 或 SOC，LT6711GXE 的 I2C/GPIO 已经连接到 MCU 或 SOC，DC-DC 芯片的使能引脚也应该连接到 MCU 或 SOC，上电顺序应该由 MCU 或 SOC 控制。

LT6711GXE 不考虑下电时序。

(If you have external MCU or SOC, I2C/GPIO of LT6711GXE have already been connected to MCU or SOC, the enable pin of DC-DC chip should be connected to MCU or SOC too, Power-on and power-off sequence should be controlled by MCU or SOC;

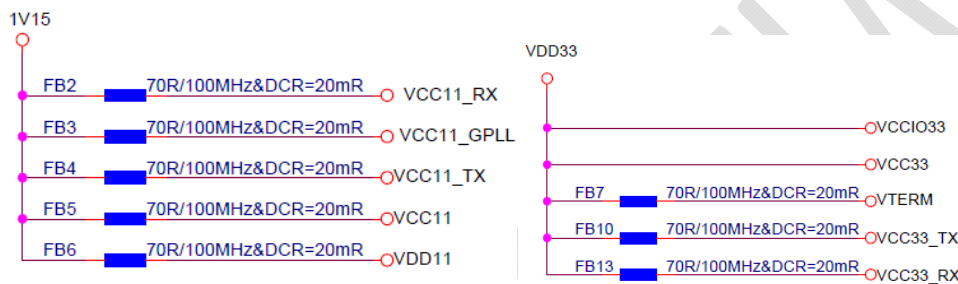
LT6711GXE do not care of the power-off sequence.)



### 3.13 Bead

电源请串联磁珠，用于抑制电源上的高频噪声，尤其是数据传输速率较高的时候，不建议减少或删除。磁珠的 DCR 请小于 20mohm，额定电流大于 1A。

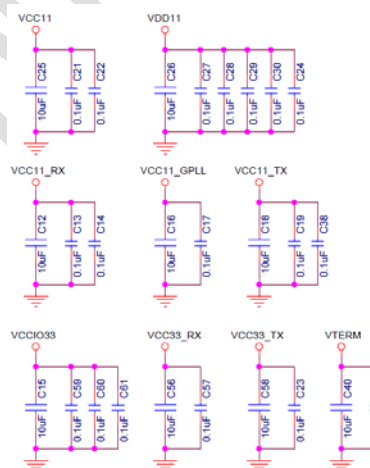
(Please connect magnetic beads in series with the power supply to suppress high-frequency noise on the power supply, especially when the data transmission rate is high, it is not recommended to reduce or delete it. The DCR of the magnetic beads should be less than 20mohm, and the rated current should be greater than 1A.)



### 3.14 Decoupling Capacitor

请尽量保证芯片每个电源 Pin 都至少有一颗 0.1uF 的电容，且靠近管脚放置。同时请在每路磁珠输出端添加一颗 10uF 去耦电容。

(Please try to make sure that each power pin of the chip has at least one 0.1uF capacitor and placed close to the pin. Please also add a 10uF decoupling capacitor to each bead output.)



## 4 PCB LAYOUT

### 4.1 Differential Impedance

TMDS 和 Main link 信号线阻抗是 100 欧姆，可以使用软件 SI9000 计算阻抗，线宽和线距也可以使用 PCB 板厂推荐值，可以要求 PCB 工厂做阻抗控制。

(TMDS and Main link signal line impedance is 100 ohms, you can use the software SI9000 to calculate the impedance, the line width and the line spacing can also be used to PCB board factory recommended values, you can ask the PCB factory to do impedance control.)

### 4.2 Differential Pair

TMDS 差分对内长度差小于 6mil，差分对之间长度差小于 400mil，总长最好控制在 2000mil 内。

DP 差分对内长度差小于 6mil，差分对间长度差小于 400mil，总长最好控制在 2000mil 内。

(The length difference within TMDS differential pairs is less than 6mil, the length difference between differential pairs is less than 400mil, and the total length is best controlled within 2000mil.)

DP The length difference within a differential pair is less than 6mil, the length difference between differential pairs is less than 400mil, and the total length is best controlled within 2000mil. )

差分对之间间距满足 3W 原则，差

分信号最外层与 GND 铜箔之间间距不小于 3W，差分对最外围包地打 GND 孔，差分对之间若空间足够可以单独包地。

差分对请保持和其它高速信号线或大电流电源线的距离。

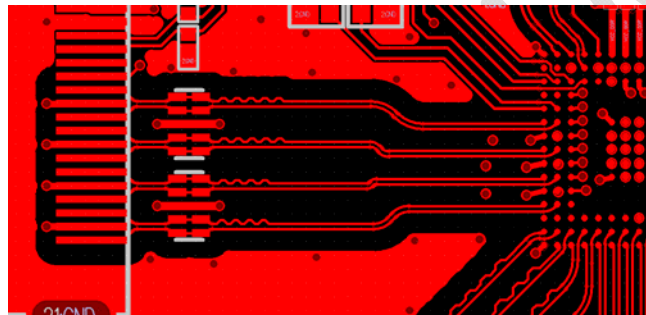
差分信号线尽量在同一层布线。同时建议将信号 Vias 的数量保持在最低限度，建议高速信号 Vias 的数量不超过 2 个。信号过孔旁边可以增加对称 GND 孔，提供信号回流路径。

(The spacing between differential pairs should meet the 3W principle, the spacing between the outermost layer of the differential signal and the GND copper foil should not

be less than 3W, the outermost periphery of the differential pairs should be wrapped with GND holes, and the differential pairs can be wrapped with ground individually if there is enough space between the pairs.

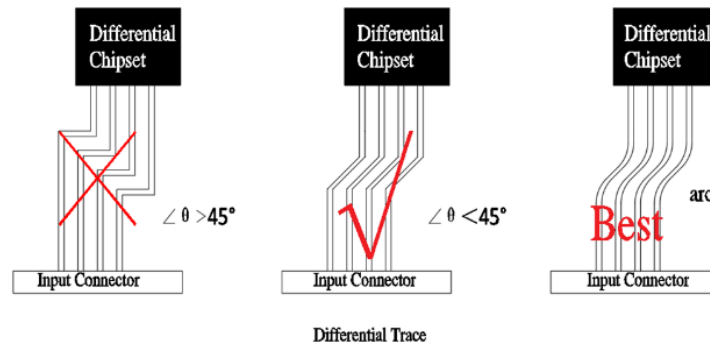
Please keep the distance between the differential pair and other high-speed signal lines or high-current power lines.

Differential signal lines should be wired in the same layer as far as possible. It is also recommended to keep the number of signal vias to a minimum. It is recommended that the number of high-speed signal vias should not exceed two. Symmetrical GND holes can be added next to the signal vias to provide a signal return path.)



差分信号用 Arc 的方式做转角。

(The differential signal uses Arc to make the corner.)



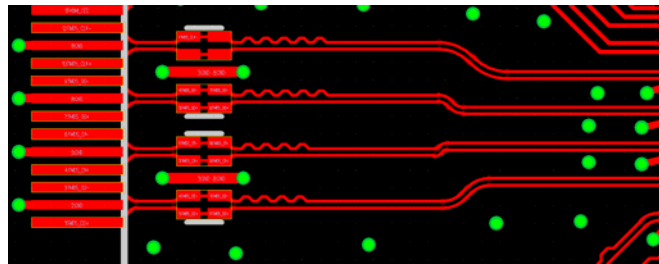
### 4.3 Reference Layer

差分对走线最好有完整的参考层，尽量不要分割参考层，信号参考层最好是 GND 层。

为了提高阻抗，HD\_DVI 连接器焊盘下方以及 EMI 器件下方的第 2 层铜箔应该被挖空，使用第三层对应位置作为参考层，同时第三层对应位置最好是 GND 层。

(Differential pair alignments should preferably have a complete reference layer, try not to split the reference layer, and the signal reference layer should preferably be the GND layer.

In order to increase the impedance, the layer 2 copper foil below the HD\_DVI connector pads and below the EMI devices should be hollowed out, using the third corresponding position as the reference layer, while the third corresponding position is preferably the GND layer.)



#### 4.4 EMI

若设备有 EMI 测试要求，建议在差分信号上增加共模电感，尤其是时钟信号，布线可以走中间层，且时钟信号上下层和旁边都有 GND 屏蔽。同时可以在芯片及高速信号走线上方增加屏蔽罩等。

若设备需要进行 ESD 测试，建议添加外部 ESD 器件，器件选型请根据自身需求进行选择。

(If the device has EMI test requirements, it is recommended to increase the common mode inductance on the differential signals, especially clock signals, wiring can go to the middle layer, and clock signals on the lower layer and next to the GND shield. At the same time in the chip and high-speed signals above the line to increase the shielding cover and so on.

If the device requires ESD testing, it is recommended to add external ESD devices, device selection, please select according to their own needs.)

#### 4.5 Power Layer

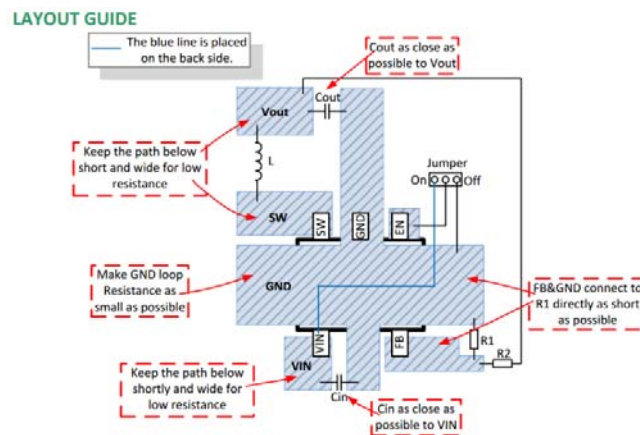
如果采用 4 层板，建议第三层走电源层，可以大面积铺铜的方法，提高电源的过流能力，同时提高散热和加大回流路径。

(If a 4-layer board is used, the third layer can be used to as power layer, and the copper can be spread over a large area to improve the overcurrent capability of the power supply, while improving heat dissipation and increasing the current return path.)

## 4.6 DC-DC Power Supply

布线方式请参考 DC- DC datasheet，注意 DC-DC 模块尽量远离高速信号，并打地孔隔离。

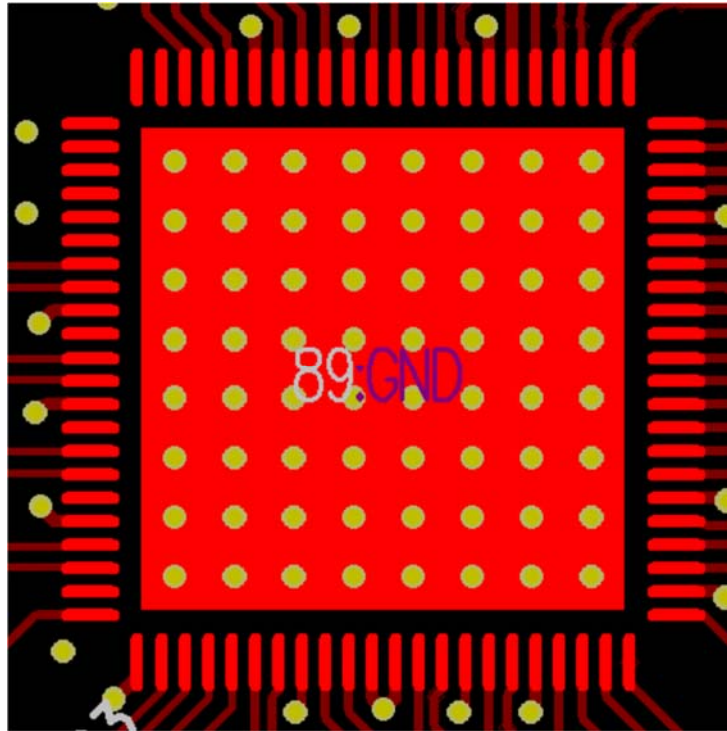
(Please refer to the DC-DC chip datasheet for the PCB layout. Note that the DC-DC chip should be kept away from high-speed signals as much as possible. PCB should be added ground via to isolate them.)



## 4.7 Signal Via

LT6711GXE 的 EPAD (89Pin) 为芯片的主要接地引脚，在 Layout 时需要在 EPAD 上均匀的打上 GND 孔，可以增加连接性和散热。

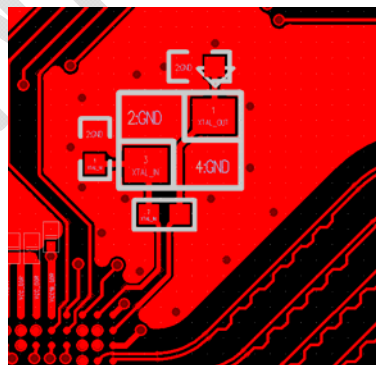
(The EPAD (89Pin) of LT6711GXE is the main ground pin of the chip. When Layout, GND holes need to be evenly punched on the EPAD to increase connectivity and heat dissipation.)



## 4.8 Crystal

如下图所示，晶振可以视为一个强干扰源，在空间允许的情况下，尽量做到远离其它高速信号，并打 GND 孔包地。

(As shown in the figure below, the crystal oscillator can be regarded as a strong interference source. When the space of PCB permits, try to keep away from other high-speed signals, and add the shield ground and ground via outside of the crystal.)



## 4.9 DDC

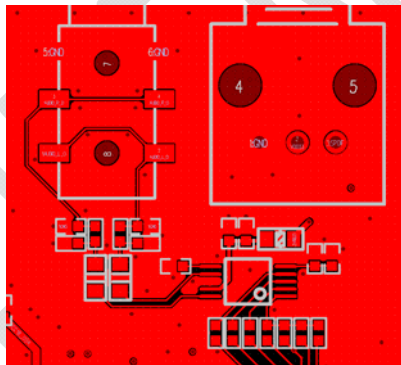
DDC 信号速率并不高，但是 DDC 信号在我们协议沟通的过程中非常重要，所以在 layout 的时候，在空间允许的情况下，尽量远离其它高速信号，做 3W（线与线之间的距离保持 3 倍线宽）。

(The DDC signal rate is not high, but the DDC signal is very important in the process of our protocol communication, so when laying out the PCB, try to keep away from other high-speed signals ,and keep the space between the other signal lines is 3 times in line width.)

## 4.10 Audio Signal

音频模拟信号尽量远离其它干扰，空间允许的情况下，走线包地，并打地孔。

(The analog signal of audio should be kept away from other interference signals, and the signal lines should be added shield ground and ground via on both sides.)



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